(ii) To elect a leader and an assistant leader

In the Fig. 7.7, the **red** shaded boxes show that same member comes twice. As, one person cannot have two leadership, therefore, the red shaded boxes cannot be counted. So there are only **12** **((4 × 1 × 4) - 4) different ways** (shown in yellow boxes and green boxes) to choose a leader and an assistant leader for the math club.

LEADER \ ASSISTANT LEADER		M	A	T	H
	M	MM	MA	MT	MH
	A	AM	AA	AT	AH
	T	TM	TA	TT	TH
	H	HM	HA	HT	HH

(1 (M or A or T or H) × 4 (Leader) × 4 (Assistant Leader)) – 4 (Leader and Assistant Leader) = 12 ways

Fig. 7.7

Activity

1. Determine the number of two digit numbers that can be formed using the digits 1, 3 and 5 with repetition of digits allowed.

The activity consists of two parts

(i) Choose a one's digit.

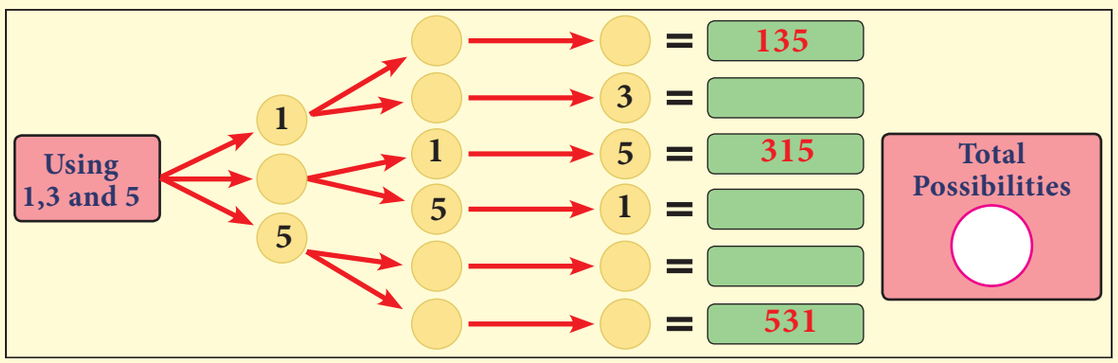
(ii) Choose a ten's digit.

Complete the table given beside

Ten's Digit \ One's Digit	1	3	5
1			15
3		33	
5	51		

2. Find the three digit numbers that can be formed using the digits 1, 3 and 5 without repetition of digits.

Complete the tree diagram given below to the numbers



DO YOU KNOW?

A **password** using 6 characters is created where the first 2 characters are any of the **alphabets**, the third character is any one special character **like @, #, $, %, &, _,+,~, * or -** and the last 3 characters are any of the numbers from **0 to 9**. For that, there are **26 × 26 × 10 × 10 × 10 × 10 = 67,60,000** number of different ways possible to create that password.

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